Ex. (3) In figure 3.75, two circles intersect each other in points X and Y. Tangents drawn from a point M on line XY touch the circles at P and Q.

Prove that, seg PM ≅ seg QM.

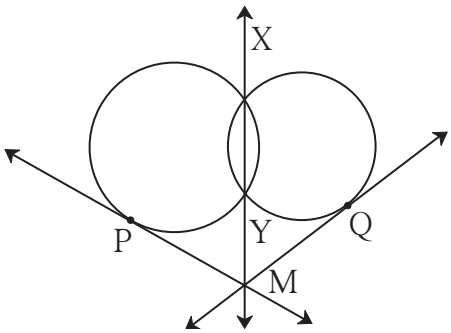


Fig. 3.75

Solution : Fill in the blanks and write proof.

Line MX is a common ..... of the two circles.

$\therefore PM^2 = MY \times MX$ ..... (I)

Similarly ..... = ... × ..... , tangent secant segment theorem .....(II)

∴ From (I) and (II) ..... = QM^2

∴ PM = QM

seg PM ≅ seg QM

Ex. (4)

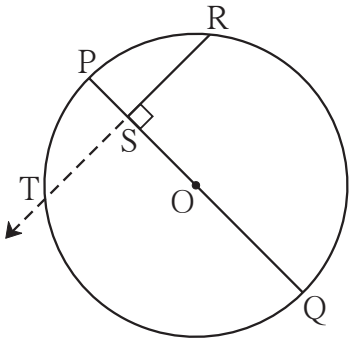


Fig. 3.76

In figure 3.76, seg PQ is a diameter of a circle with centre O. R is any point on the circle.

seg RS ⊥ seg PQ.

Prove that, SR is the geometric mean of PS and SQ.

[That is, $SR^2 = PS \times SQ$]

Solution : Write the proof with the help of the following steps.

(1) Draw ray RS. It intersects the circle at T.

(2) Show that RS = TS.

(3) Write a result using theorem of intersection of chords inside the circle.

(4) Using RS = TS complete the proof.



Let's think.

(1) In figure 3.76, if seg PR and seg RQ are drawn, what is the nature of Δ PRQ ?

(2) Have you previously proved the property proved in example (4) ?

Practice set 3.5

1. In figure 3.77, ray PQ touches the circle at point Q. PQ = 12, PR = 8, find PS and RS.

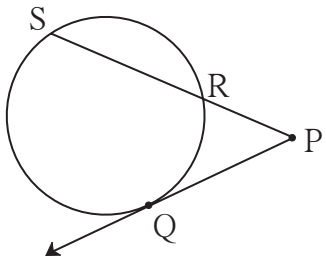


Fig. 3.77

2. In figure 3.78, chord MN and chord RS intersect at point D.

(1) If RD = 15, DS = 4, MD = 8 find DN

(2) If RS = 18, MD = 9, DN = 8 find DS

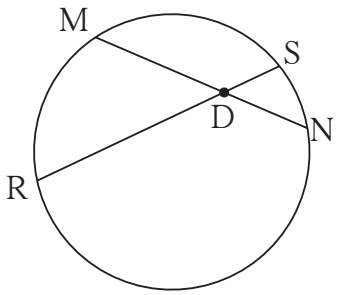


Fig. 3.78

3. In figure 3.79, O is the centre of the circle and B is a point of contact. seg OE ⊥ seg AD, AB = 12, AC = 8, find

(1) AD (2) DC (3) DE.

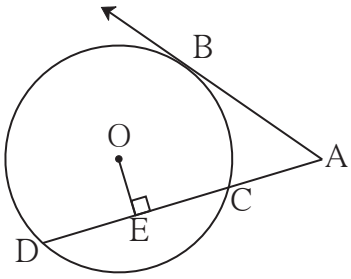


Fig. 3.79

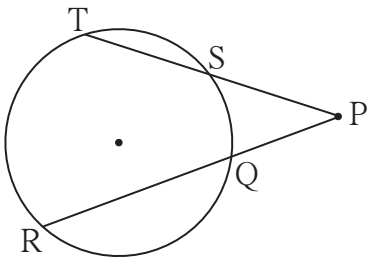


Fig. 3.80

4. In figure 3.80, if PQ = 6, QR = 10, PS = 8 find TS.

5. In figure 3.81, seg EF is a diameter and seg DF is a tangent segment. The radius of the circle is r. Prove that, DE × GE = $4r^2$

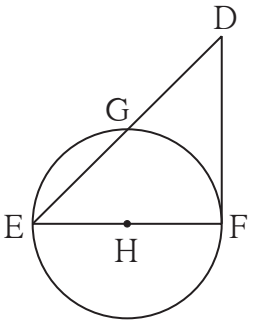


Fig. 3.81

Problem set 3

1. Four alternative answers for each of the following questions are given. Choose the correct alternative.

(1) Two circles of radii 5.5 cm and 3.3 cm respectively touch each other. What is the distance between their centers ?

(A) 4.4 cm (B) 8.8 cm (C) 2.2 cm (D) 8.8 or 2.2 cm

(2) Two circles intersect each other such that each circle passes through the centre of the other. If the distance between their centres is 12, what is the radius of each circle ?

(A) 6 cm (B) 12 cm (C) 24 cm (D) can't say

(3) A circle touches all sides of a parallelogram. So the parallelogram must be a, ...

(A) rectangle (B) rhombus (C) square (D) trapezium

(4) Length of a tangent segment drawn from a point which is at a distance 12.5 cm from the centre of a circle is 12 cm, find the diameter of the circle.

(A) 25 cm (B) 24 cm (C) 7 cm (D) 14 cm

(5) If two circles are touching externally, how many common tangents of them can be drawn?

(A) One (B) Two (C) Three (D) Four

(6) $\angle$ACB is inscribed in arc ACB of a circle with centre O. If $\angle ACB = 65°$, find m(arc ACB).

(A) $65°$ (B) $130°$ (C) $295°$ (D) $230°$

(7) Chords AB and CD of a circle intersect inside the circle at point E. If AE = 5.6, EB = 10, CE = 8, find ED.

(A) 7 (B) 8 (C) 11.2 (D) 9

(8) In a cyclic $\square$ABCD, twice the measure of $\angle$ A is thrice the measure of $\angle$C. Find the measure of $\angle$C ?

(A) 36 (B) 72 (C) 90 (D) 108

9)* Points A, B, C are on a circle, such that m(arc AB) = m(arc BC) = $120°$. No point, except point B, is common to the arcs. Which is the type of Δ ABC ?

(A) Equilateral triangle (B) Scalene triangle

(C) Right angled triangle (D) Isosceles triangle

(10) Seg XZ is a diameter of a circle. Point Y lies in its interior. How many of the following statements are true ?

(i) It is not possible that $\angle$ XYZ is an acute angle.

(ii) $\angle$ XYZ can't be a right angle.

(iii) $\angle$ XYZ is an obtuse angle.

(iv) Can't make a definite statement for measure of $\angle$ XYZ.

(A) Only one (B) Only two (C) Only three (D) All

2. Line l touches a circle with centre O at point P. If radius of the circle is 9 cm, answer the following.

(1) What is d(O, P) = ? Why ?

(2) If d(O, Q) = 8 cm, where does the point Q lie ?

(3) If d(PQ) = 15 cm, How many locations of point R are line on line l ? At what distance will each of them be from point P ?

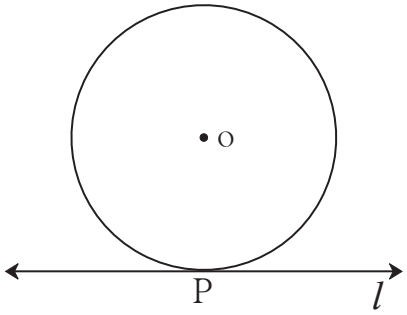


Fig. 3.82

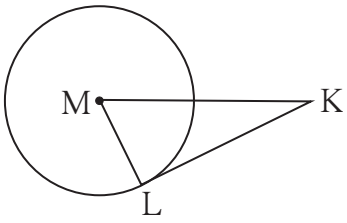


Fig. 3.83

3. In figure 3.83, M is the centre of the circle and seg KL is a tangent segment.

If MK = 12, KL = $6\sqrt{3}$ then find -

(1) Radius of the circle.

(2) Measures of $\angle$ K and $\angle$ M.

4. In figure 3.84, O is the centre of the circle. Seg AB, seg AC are tangent segments. Radius of the circle is r and l(AB) = r, Prove that, $\square$ABOC is a square.

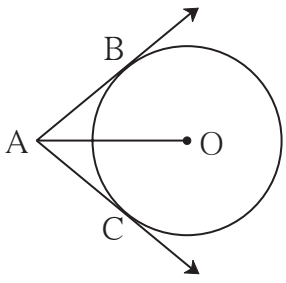


Fig. 3.84

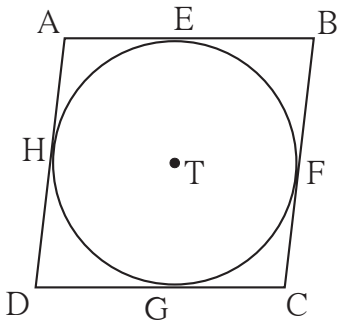


Fig. 3.85

5. In figure 3.85, □ABCD is a parallelogram. It circumscribes the circle with cnetre T. Point E, F, G, H are touching points. If AE = 4.5, EB = 5.5, find AD.

6. In figure 3.86, circle with centre M touches the circle with centre N at point T. Radius RM touches the smaller circle at S. Radii of circles are 9 cm and 2.5 cm. Find the answers to the following questions hence find the ratio MS:SR.

(1) Find the length of segment MT

(2) Find the length of seg MN

(3) Find the measure of ∠ NSM.

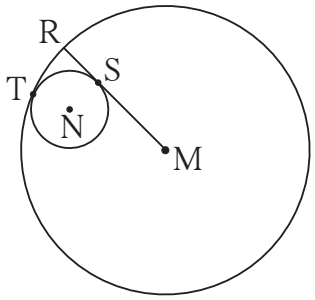


Fig. 3.86

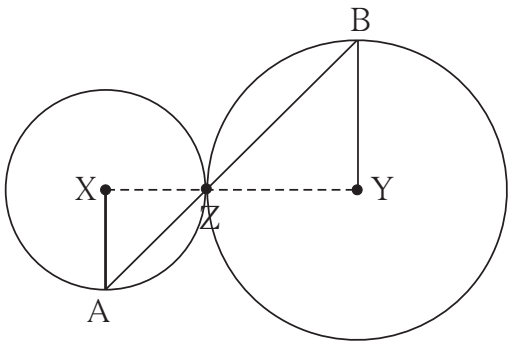


Fig. 3.87

7. In the adjoining figure circles with centres X and Y touch each other at point Z. A secant passing through Z intersects the circles at points A and B respectively. Prove that,

radius XA || radius YB.

Fill in the blanks and complete the proof.

Construction : Draw segments XZ and ..... .

Proof :By theorem of touching circles, points X, Z, Y are ..... .

∴ ∠ XZA ≅ ..... opposite angles

Let ∠ XZA = ∠ BZY = *a* ..... (I)

Now, seg XA ≅ seg XZ ..... (.....)

∴ ∠ XAZ = ... = *a* ..... (isosceles triangle theorem) (II)

similarly, seg YB ≅ .... (.....)

∴ ∠ BZY = ... = *a* ..... (.....) (III)

$\therefore$ from (I), (II), (III),

$\angle$ XAZ = ...

$\therefore$ radius XA || radius YB ..... (.....)

8. In figure 3.88, circles with centres X and Y touch internally at point Z . Seg BZ is a chord of bigger circle and it itersects smaller circle at point A.
Prove that, seg AX || seg BY.

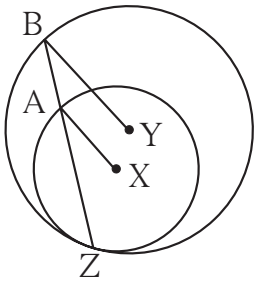


Fig. 3.88

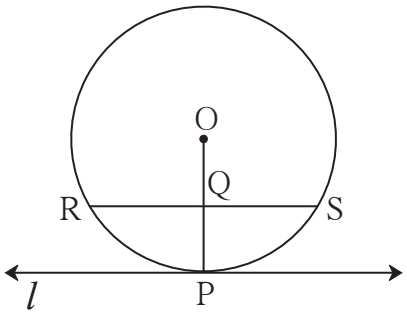


Fig. 3.89

9. In figure 3.89, line l touches the circle with centre O at point P. Q is the mid point of radius OP. RS is a chord through Q such that chords RS || line l. If RS = 12 find the radius of the circle.

10*. In figure 3.90, seg AB is a diameter of a circle with centre C. Line PQ is a tangent, which touches the circle at point T.
seg AP $\perp$ line PQ and seg BQ $\perp$ line PQ.
Prove that, seg CP $\cong$ seg CQ.

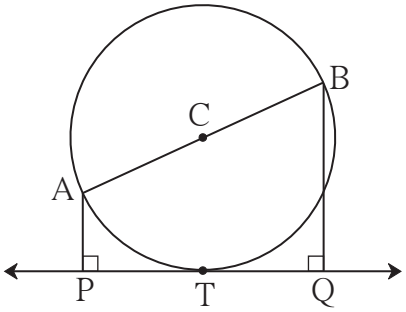


Fig. 3.90

11*. Draw circles with centres A, B and C each of radius 3 cm, such that each circle touches the other two circles.

12*. Prove that any three points on a circle cannot be collinear.

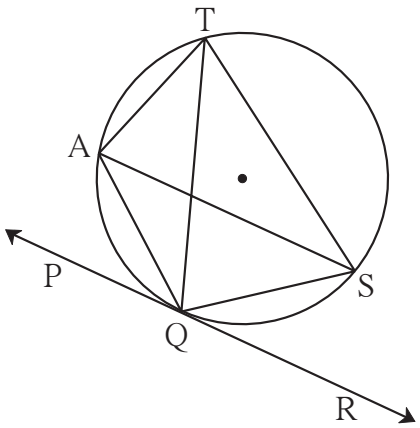


Fig. 3.91

13. In figure 3.91, line PR touches the circle at point Q. Answer the following questions with the help of the figure.

(1) What is the sum of ∠TAQ and ∠TSQ ?

(2) Find the angles which are congruent to ∠AQP.

(3) Which angles are congruent to ∠QTS ?

(4) ∠TAS = 65°, find the measure of ∠TQS and arc TS.

(5) If ∠AQP = 42°and ∠SQR = 58° find measure of ∠ATS.

14. In figure 3.92, O is the centre of a circle, chord PQ ≅ chord RS

If ∠ POR = 70°

and (arc RS) = 80°, find –

(1) m(arc PR)

(2) m(arc QS)

(3) m(arc QSR)

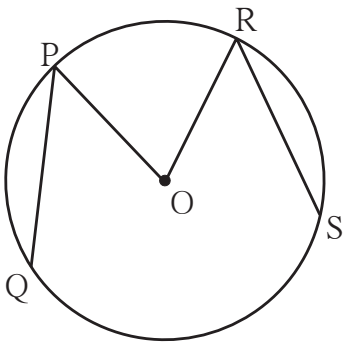


Fig. 3.92

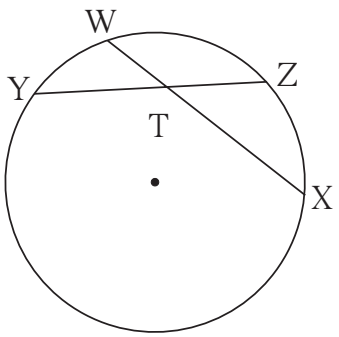


Fig. 3.93

15. In figure 3.93, m(arc WY) = 44°, m(arc ZX) = 68°, then

(1) Find the measure of ∠ ZTX.

(2) If WT = 4.8, TX = 8.0, YT = 6.4, find TZ.

(3) If WX = 25, YT = 8, YZ = 26, find WT.

16. In figure 3.94,

(1) m(arc CE) = 54°,

m(arc BD) = 23°, find measure of ∠CAE.

(2) If AB = 4.2, BC = 5.4,

AE = 12.0, find AD

(3) If AB = 3.6, AC = 9.0,

AD = 5.4, find AE

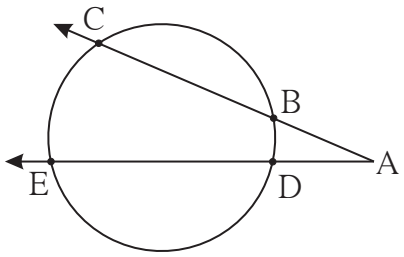


Fig. 3.94

17. In figure 3.95, chord EF || chord GH. Prove that, chord EG ≅ chord FH.

Fill in the blanks and write the proof.

Proof : Draw seg GF.

∠ EFG = ∠ FGH ..... ☐ (I)

∠ EFG = ☐ ..... inscribed angle theorem (II)

∠ FGH = ☐ ..... inscribed angle theorem (III)

∴ m(arc EG) = ☐ from (I), (II), (III).

chord EG ≅ chord FH ..... ☐

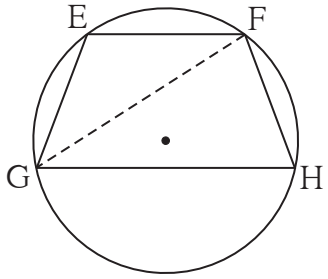


Fig. 3.95

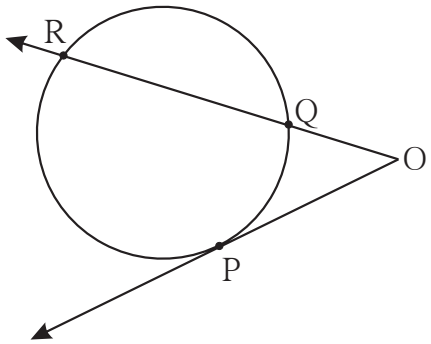


Fig. 3.96

18. In figure 3.96 P is the point of contact.

(1) If m(arc PR) = 140°,

∠ POR = 36°,

find m(arc PQ)

(2) If OP = 7.2, OQ = 3.2,

find OR and QR

(3) If OP = 7.2, OR = 16.2,

find QR.

19. In figure 3.97, circles with centres C and D touch internally at point E. D lies on the inner circle. Chord EB of the outer circle intersects inner circle at point A. Prove that, seg EA ≅ seg AB.

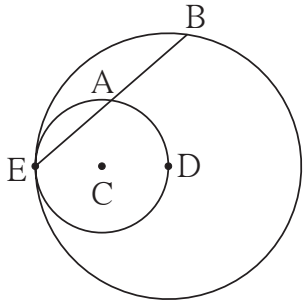


Fig. 3.97

20. In figure 3.98, seg AB is a diameter of a circle with centre O . The bisector of ∠ACB intersects the circle at point D. Prove that, seg AD ≅ seg BD.
Complete the following proof by filling in the blanks.

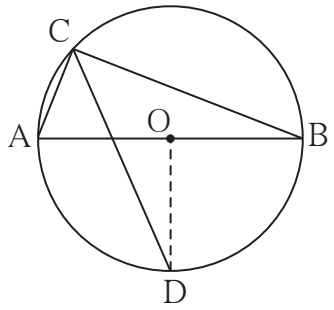


Fig. 3.98

Proof : Draw seg OD.

∠ ACB = [ ] ..... angle inscribed in semicircle

∠ DCB = [ ]..... CD is the bisector of ∠ C

m(arc DB) = [ ]..... inscribed angle theorem

∠ DOB = [ ]..... definition of measure of an arc (I)

seg OA ≅ seg OB ..... [ ] (II)

∴ line OD is [ ] of seg AB ..... From (I) and (II)

∴ seg AD ≅ seg BD

21. In figure 3.99, seg MN is a chord of a circle with centre O. MN = 25,
L is a point on chord MN such that ML = 9 and d(O,L) = 5.
Find the radius of the circle.

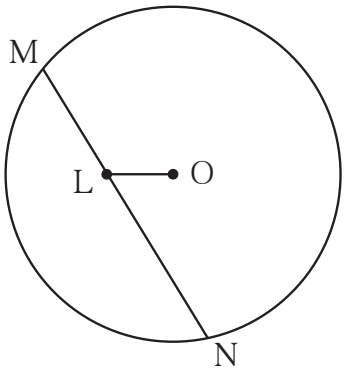


Fig. 3.99

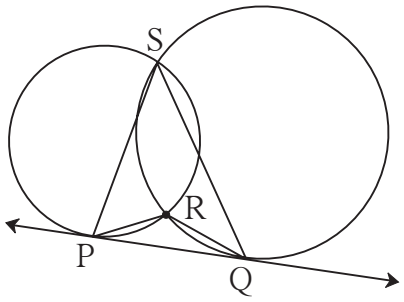


Fig. 3.100

22*. In figure 3.100, two circles intersect each other at points S and R. Their common tangent PQ touches the circle at points P, Q.
Prove that, ∠ PRQ + ∠ PSQ = 180°